CHEMISTRY

SECTION – I
Single Correct Choice Type

This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. A vessel filled with He gas at a pressure of 4 bar had a small thin orifice through which helium escaped into an evacuated space at the rate of 7.07 mmole/hr. How long would it take for 10 mmole of O_2 to leak through a similar orifice if O_2 were confined at same pressure and temperature? (Atomic mass of He = 4, O = 16)

(a) 2 hr (b) 3 hr
(c) 4 hr (d) 5 hr

2. When $NaNO_3$ (d = 2.0 g/cc) is heated in a closed vessel of 100 ml, oxygen is liberated and $NaNO_2$ (d = 1.5 g/cc) is left behind as per the reaction: $2NaNO_3(s) \rightleftharpoons 2NaNO_2(s) + O_2(g)$. At equilibrium, the volumes of $NaNO_3$ left and $NaNO_2$ produced are very small and can be neglected. Which of the following is a correct statement about this equilibrium?

(a) Addition of 30 g of $NaNO_2$ favours forward reaction.
(b) Addition of 30 g of $NaNO_3$ favours reverse reaction.
(c) Increasing temperature favours reverse reaction.
(d) None of these.

3. When 60 ml of 0.1M $Ca(NO_3)_2$ is mixed with 40 ml of 0.125M Na_2CO_3, $CaCO_3$ precipitates. If Ksp of $CaCO_3$ is 5×10^{-9} M^2, the $[CO_3^{2-}]$ in the resulting solution is

(a) 5×10^{-6} M (b) 5×10^{-7} M
(c) 5×10^{-8} M (d) 5×10^{-9} M

4. If 0.5 L of a 1 M $SnSO_4$ solution is electrolysed for a period of 30 minutes using a current of 9.65 A and inert electrodes, the final concentration of Sn^{2+} in the solution will be (Given: $E^{o}_{Sn^{2+}|Sn} = -0.14V$, $E^{o}_{Sn^{2+}|Sn^{4+}} = -0.13V$, $E^{o}_{OH^{-}|O_2} = -0.4V$, $E^{o}_{SO_4^{2-}|S_2O_8^{2-}} = -2.0V$)

(a) 0.64 M (b) 0.28 M

(c) 1.18 M (d) 0.84 M

5. The number of linkage isomers for the compound $K_2[Cu(CN)_2\ (NCS)_2]$ are (including it)

(a) 4 (b) 6

(c) 16 (d) 9

6. Which of the following is a correct statement about s-block compounds?

(a) Stability of metal chlorides decreases down the alkali metal group.

(b) Lattice energy of M_2O_2 is greater than that of M_2O.

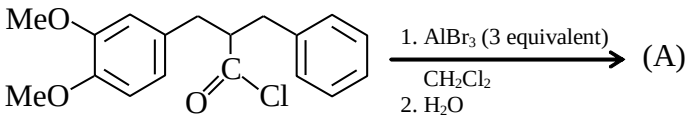
(c) Stability of peroxide increases with the increase in size of alkali metal cation.

(d) The water of crystallization is greater in alkali metal salts than in alkaline earth metal salts.

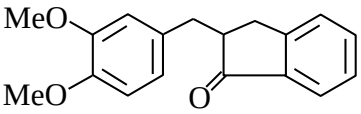
7. Amongst NO_3^-, AsO_3^{3-}, CO_3^{2-}, ClO_3^-, SO_3^{2-} and BO_3^{3-}, the non-planar species are (atomic no: N = 7, As = 33, C = 6, Cl = 17, S = 16, B = 5)

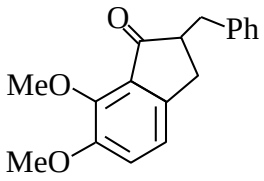
(a) CO_3^{2-}, SO_3^{2-} and BO_3^{3-} (b) AsO_3^{3-}, ClO_3^- and SO_3^{2-}

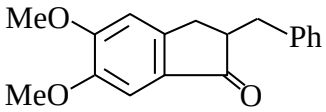
(c) NO_3^-, CO_3^{2-} and BO_3^{3-} (d) SO_3^{2-}, NO_3^- and BO_3^{3-}

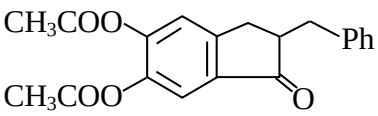
8. 

The product (A) would be

(a) 

(b) 

(c) 

(d) 

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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9. When radioactive sulfur is added to an alkaline sodium sulfite solution, radioactive thiosulphate ion is formed. Upon adding Ba^{2+}, a precipitate of BaS_2O_3 is formed. The precipitate is filtered, dried and then treated with acid, producing solid sulfur, SO_2 gas and water. The correct set of products is (* indicates radioactive sulfur)

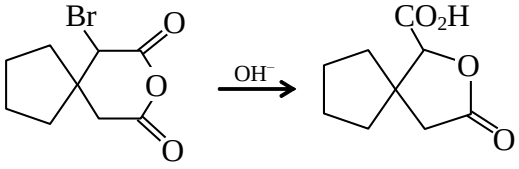
(a) $SO_2 + S_8$ (b) $SO_2 + \overset{*}{S}_8$

(c) $\overset{*}{S}O_2 + S_8$ (d) $\overset{*}{S}O_2 + \overset{*}{S}_8$

10. Which of the following oxyacid(s) of sulphur have S–S bond?

(a) $H_2S_2O_8$ (b) $H_2S_2O_4$

(c) $H_2S_2O_5$ (d) $H_2S_2O_6$

11. 

The given reaction involves:

(a) Nucleophilic substitution at acyl carbon.

(b) Nucleophilic substitution at alkyl carbon.

(c) Elimination

(d) Ring contraction

12. The amount of dissolved oxygen in 1 litre water in equilibrium with the air at 1 atm pressure at 25°C will be (assume that air contains 20 mole % oxygen, Henry's constant (k_H) for oxygen is 3.04×10^7 mm Hg and density of H_2O at 25°C is 1 g/cc)

(a) the mole fraction of O_2 in solution is 5×10^{-7}.

(b) the mole fraction of O_2 in solution is 5×10^{-6}.

(c) the molarity of O_2 in solution is 2.77×10^{-5} M

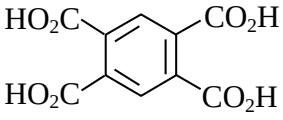
(d) the molarity of O_2 in solution is 2.77×10^{-4}M

SECTION–III
Comprehension Type

This section contains 1 group of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage–I

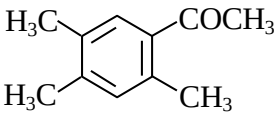
Compound (A), $C_{11}H_{12}O$, which gave no Tollen's test, was treated with $LiAlH_4$, followed by dilute acid, to give compound (B), which could be resolved into enantiomers. When optically active (B) was treated with CrO_3 in pyridine, an optically inactive sample of (A) was obtained. Heating (A) with hydrazine in base gave hydrocarbon (C), which, when heated with alkaline $KMnO_4$ followed by acidification, gave carboxylic acid (D).

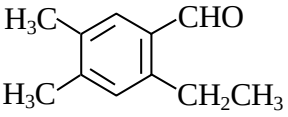


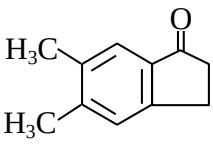
(D)

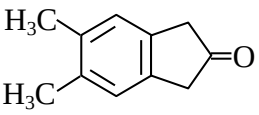
Another compound (X), $C_6H_{12}O_2$, was found to be optically active and was slowly oxidized to an optically active carboxylic acid (Y), $C_6H_{12}O_3$, by $[Ag(NH_3)_2]^+$. Oxidation of (X) by anhydrous CrO_3 gave an optically inactive compound that reacted with Zn amalgam/HCl to give 3-methylpentane. With aqueous H_2CrO_4, compound (X) was oxidized to an optically inactive dicarboxylic acid (Z), $C_6H_{10}O_4$.

13. The most probable structure of (A) will be

(a) 

(b) 

(c) 

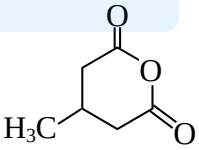
(d) 

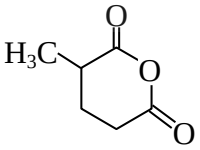
14. The structure of compound (X) would be

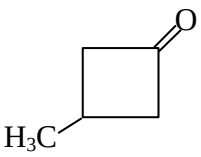
(a) $OHCCH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_2CHO$ (b) $OHCCH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–COCH_3$

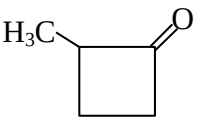
(c) $HOH_2CCH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–COCH_3$ (d) $OHCCH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_2CH_2OH$

15. Compound (Z) on heating with barium oxide forms

(a) 

(b) 

(c) 

(d) 

SECTION–IV

Matrix–Match Type

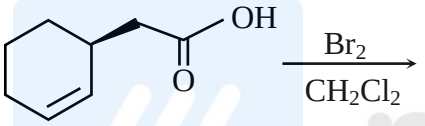
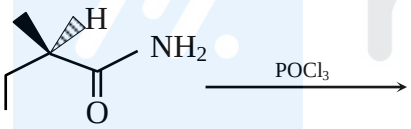
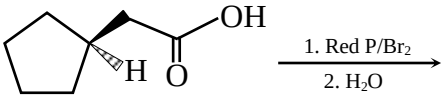
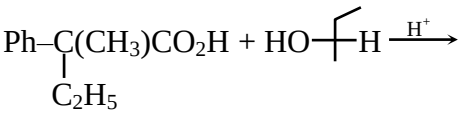
This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

	1	2	3	4	5
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

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1. Match the following

Column I	**Column II**
(A) 	**1.** Product is a racemic mixture.
(B) 	**2.** Substrate can show optical isomerism.
(C) 	**3.** Product is optically active.
(D) 	**4.** Product is capable of showing geometrical isomerism. **5.** Diastereomers are formed.

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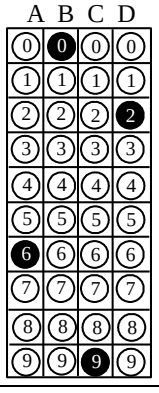
2. Match the following

Column I	**Column II**
(A) 2.0 g NaOH + 5.3 g Na_2CO_3 in 1 L H_2O	**1.** Vol. of 0.1M HCl used in presence of phenolphthalein indicator = 500 mL
(B) 1.0 g NaOH + 2.1 g $NaHCO_3$ in 1L H_2O	**2.** Vol. of 0.1 M HCl used in presence of methyl orange indicator = 500 mL
(C) 5.3 g Na_2CO_3 + 2.1 g $NaHCO_3$ in 1 L H_2O	**3.** Mass of NaCl produced in presence of phenolphthalein indicator = 5.85
(D) 1.06 g Na_2CO_3 in 1 L H_2O	**4.** Mass of NaCl produced in presence of methyl orange indicator = 1.17 g
	5. Mass of $CaCO_3$ precipitated after addition of excess of $CaCl_2$ solution to it. = 5 g.

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SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



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1. A 40 ml sample of an aqueous solution of the methylamine at 25°C is titrated with 0.15 M HCl and the equivalence point is reached when 40 ml of the acid have been added. The pH at the equivalence point is ____ (Given : K_a for $CH_3NH_3^+$ is $\frac{4}{3}\times10^{-11}$).

Space for work

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2. The number of $R_2Si(OH)_2$ units required to prepare a silicone polymer containing 10 $Si-O-Si$ linkages is ——.

Space for work

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3. A compound with molecular formula $C_6H_{14}O_5$ is converted by the action of acetyl chloride to a compound with molecular mass 334. The number of OH groups in the compound is ——.

Space for work

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4. The ground state electronic wave function of hydrogen atom is proportional to e^{-r/a_0} where r is the distance from the nucleus and a_0 is a constant.

Calculate the relative probability of finding the electron inside a region of volume 1.0 pm^3, located at the nucleus to a distance of a_0 from the nucleus. Report your answer in single digit ignoring any fraction involved. (Given: ln 0.14 = – 2)

Space for work

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